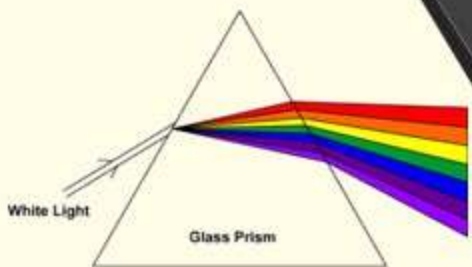


PHYSICS

WORK & ENERGY

MODULE - 11



WHAT DOES WORK MEAN?



A HUGE WORKLOAD



A HEAVY WORKOUT



DOING NO WORK

How is this different from Physics specifically?

WORK- MEANING IN PHYSICS

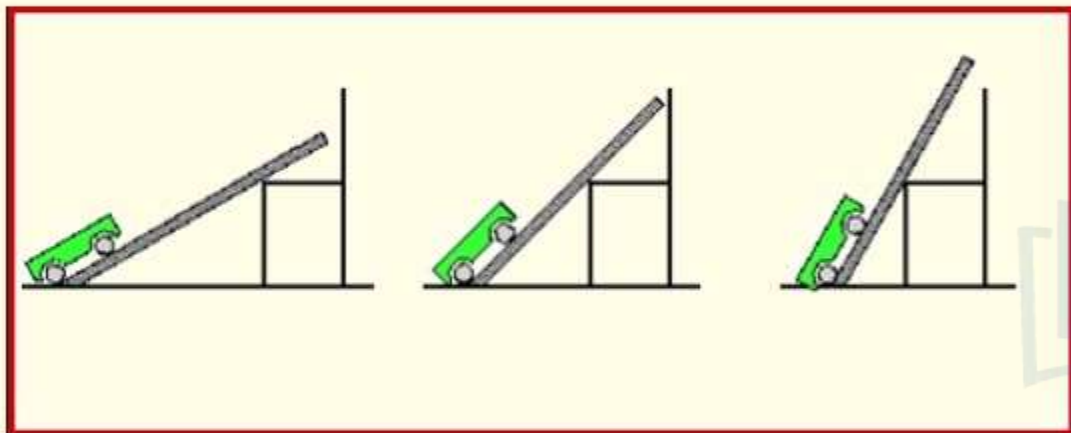


- *Have we applied any force?(any unbalanced force)*
- *Is the object moving parallel to the applied force?*

When an object moves parallel to the applied force, it is called work done on that object.

Work is a scalar quantity.

WORK – MATHEMATICAL EXPRESSION



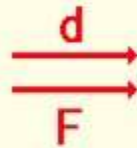
As the inclination or θ increases, more Force is required to do the work of pulling the cart against the slope.

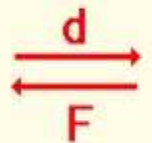
- $W = F \times D \cos \theta$
- F is the force,
- d is the displacement, and
- the angle (theta) is defined as the angle between the force and the displacement vector. It is a specific angle.
- The SI unit is Joule
- CGS unit is ergs.
- $1 \text{ Joule} = 10^7 \text{ ergs.}$





CHANGE OF ANGLE BETWEEN FORCE AND DISPLACEMENT


 $\Theta = 0 \text{ degrees}$


 $\Theta = 180 \text{ degrees}$


 $\Theta = 90 \text{ degrees}$

$$\begin{aligned}\cos 0^\circ &= 1 \\ \cos 180^\circ &= -1, \\ \cos 90^\circ &= 0\end{aligned}$$

SCENARIO A

A force acts rightward upon an object as it is displaced rightward. In such an instance, the force vector and the displacement vector are in the same direction. Thus, the angle between F and d is 0 degrees.

SCENARIO B

A force acts leftward upon an object that is displaced rightward. In such an instance, the force vector and the displacement vector are in the opposite direction. Thus, the angle between F and d is 180 degrees.

SCENARIO C

A force acts upward on an object as it is displaced rightward. In such an instance, the force vector and the displacement vector are at right angles to each other. Thus, the angle between F and d is 90 degrees.



WORK

POSITIVE WORK



$$\begin{array}{c} \xrightarrow{d} \\ \xrightarrow{F} \end{array} \quad \Theta = 0 \text{ degrees}$$

NEGATIVE WORK



$$\begin{array}{c} \xrightarrow{d} \\ \xleftarrow{F} \end{array} \quad \Theta = 180 \text{ degrees}$$

ZERO WORK

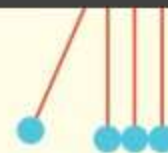


$$\begin{array}{c} \xrightarrow{d} \\ \uparrow F \end{array} \quad \Theta = 90 \text{ degrees}$$



POSITIVE WORK

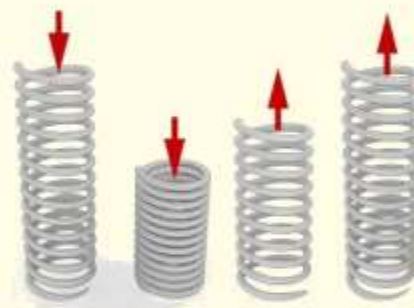
*When the angle is below 90°
the work done is positive*



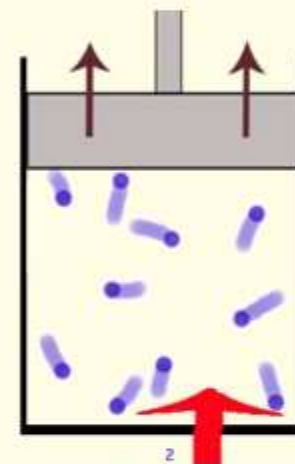
When a body falls freely under gravity



A lawn roller is pushed along the handle



A spring is stretched.



A gas cylinder with a moving piston is allowed to expand.



NEGATIVE WORK

When the angle is above 90° the work done is negative



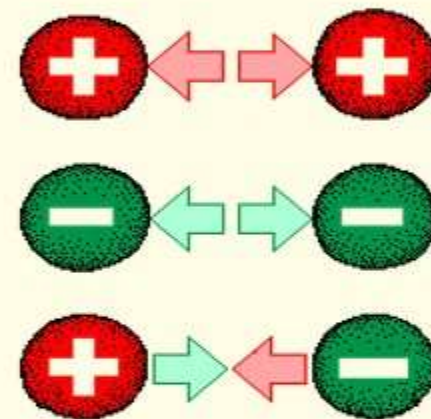
When a body is thrown up, its motion is opposed by gravity, angle between F and d is 180°



When brakes are applied to a moving vehicle, the body moves opposing the force.



A body is moved over a rough horizontal surface, the motion is opposed by frictional force.



When two like charges are brought together

ZERO WORK

When the angle between force and displacement is right angle



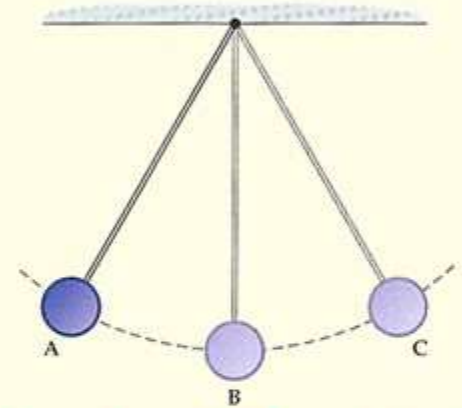
When we fail to move a heavy stone.



Coolie carrying a load in head on a horizontal road/platform



Person with a briefcase moves on a horizontal road



Tension in the string and the bob of the pendulum

The coolie will do positive work when he lifts the loads against gravitational force and takes them over his head.



WHAT DOES ENERGY MEAN?



HIGH ENERGY



LOW ENERGY

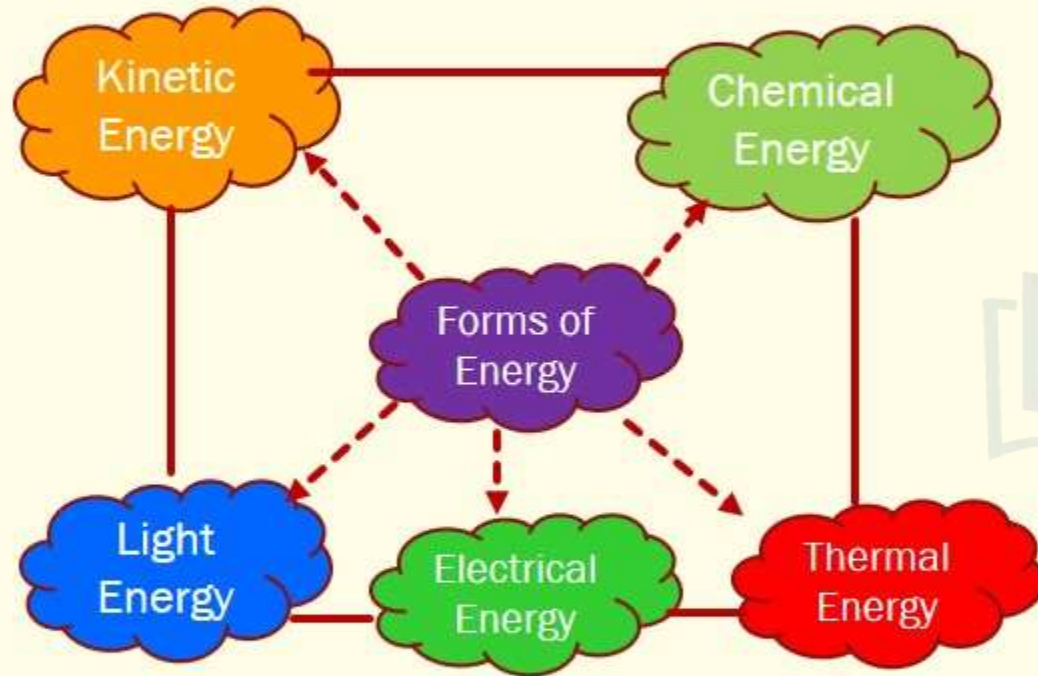


GOOD/ EVIL ENERGY



Learning Space

ENERGY



Energy is the capacity of a physical system to perform work. Energy exists in several forms such as

- Thermal/heat energy
- Mechanical energy
- Electrical energy
- Chemical energy
- Solar energy

Energy is also a scalar quantity and can be measured in Joules





MECHANICAL ENERGY

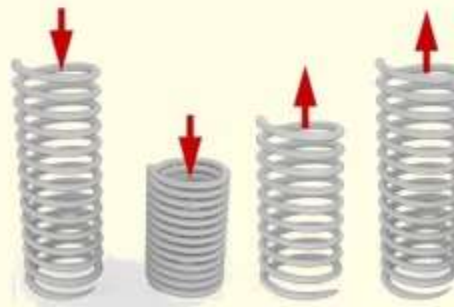
POTENTIAL ENERGY

KINETIC ENERGY

GRAVITATIONAL POTENTIAL ENERGY



ELASTIC POTENTIAL ENERGY



POTENTIAL ENERGY

GRAVITATIONAL POTENTIAL ENERGY



Energy possessed by a body by virtue of its position.

It depends on

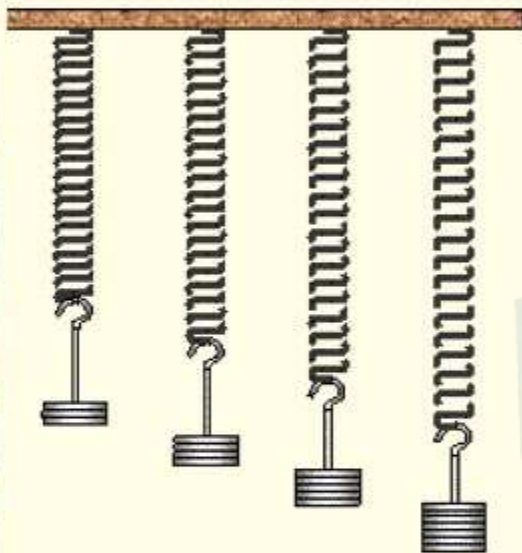
- Mass of the body
- The acceleration due to gravity
- The height or its position from the ground level.

Since many of the research work is done on table tops, the table is considered the ground level



POTENTIAL ENERGY

ELASTIC POTENTIAL ENERGY



- Elastic potential energy is the energy stored in elastic materials as the result of their stretching or compressing.
- Elastic potential energy can be stored in
 - rubber bands,
 - bungee chords,
 - trampolines,
 - springs,
 - an arrow drawn into a bow, etc.
- The amount of elastic potential energy stored in such a device is related to the amount of stretch of the device - **the more stretch, the more stored energy.**
- the more compression there is, the more force that is required to compress it further.

$$P.E = K \cdot x$$

K = spring constant

x = the amount of compression



**QUESTION 1**

When air bubble rises up in water, the Potential energy of the water

- a) Decreases
- b) Increases
- c) Remains the same
- d) Becomes doubled

ANSWER : The potential energy of the gas inside the bubble increases because the gas has moved upwards.

The potential energy of the water decreases because some water moves downwards to 'make space' for the upwards-moving gas.



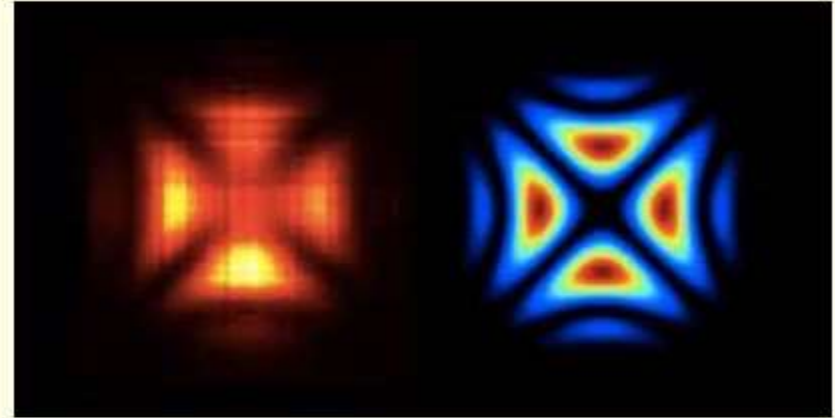
**QUESTION 2**

Work is done by us when two protons are brought towards each other, thus the potential energy

- a) Decreases
- b) Increases
- c) Remains the same
- d) Becomes doubled

ANSWER : A repulsive force acts between protons, if they are brought nearer, work must be done by external force.

Hence the potential energy of system increases.



KINETIC ENERGY



$$K.E = \frac{1}{2} m.v^2$$

m = mass of the object

V = the velocity of the object

- The kinetic energy is the energy of the motion
- Like motion, the kinetic energy can be divided as
 - Translational Kinetic energy
 - Rotational Kinetic energy
 - Vibrational Kinetic energy

Kinetic energy increases as a square product of the increase of the velocity. But it increases linearly with the mass.



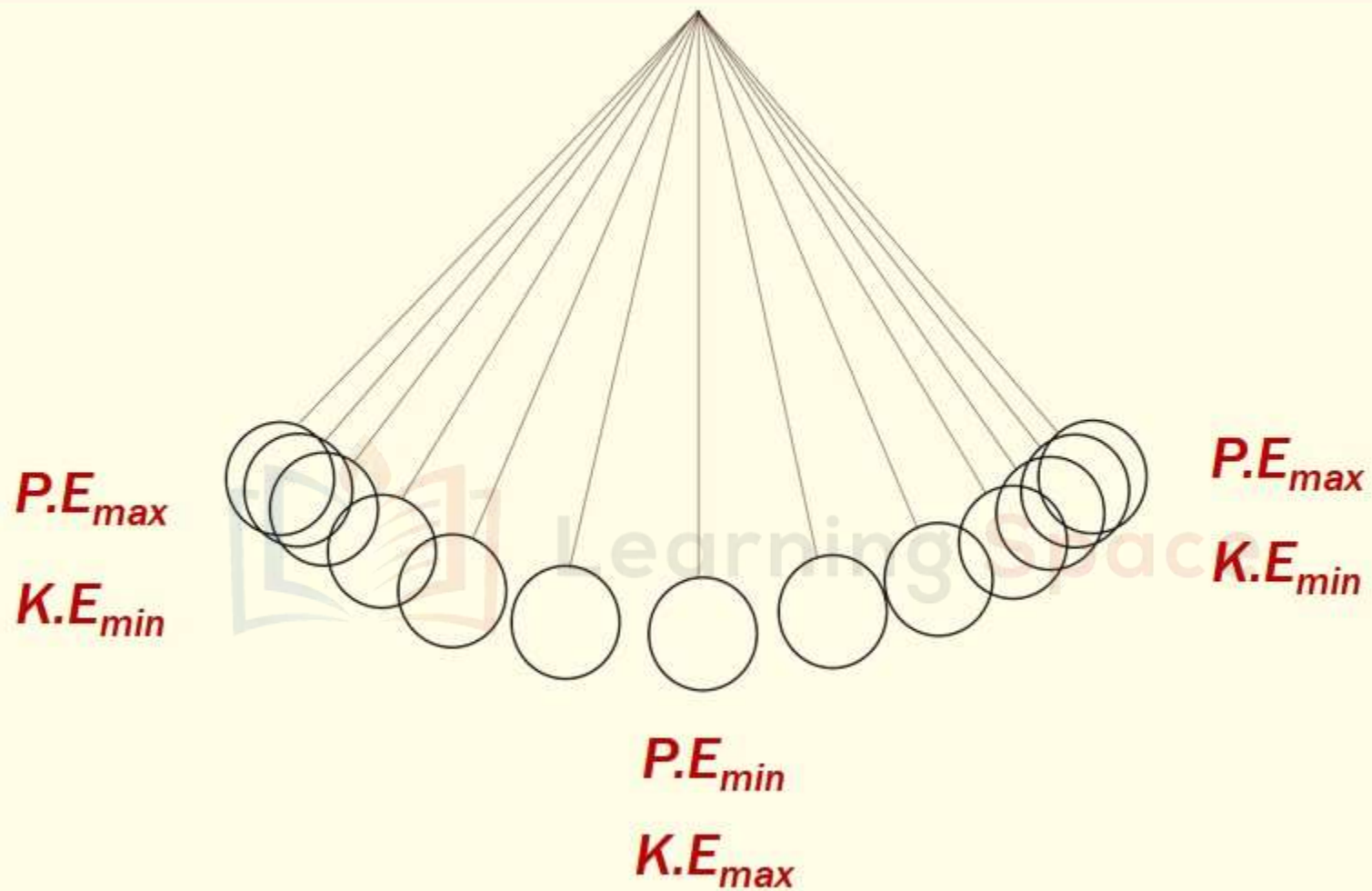
**QUESTION 3**

What happens to the kinetic energy when the velocity of the object doubles

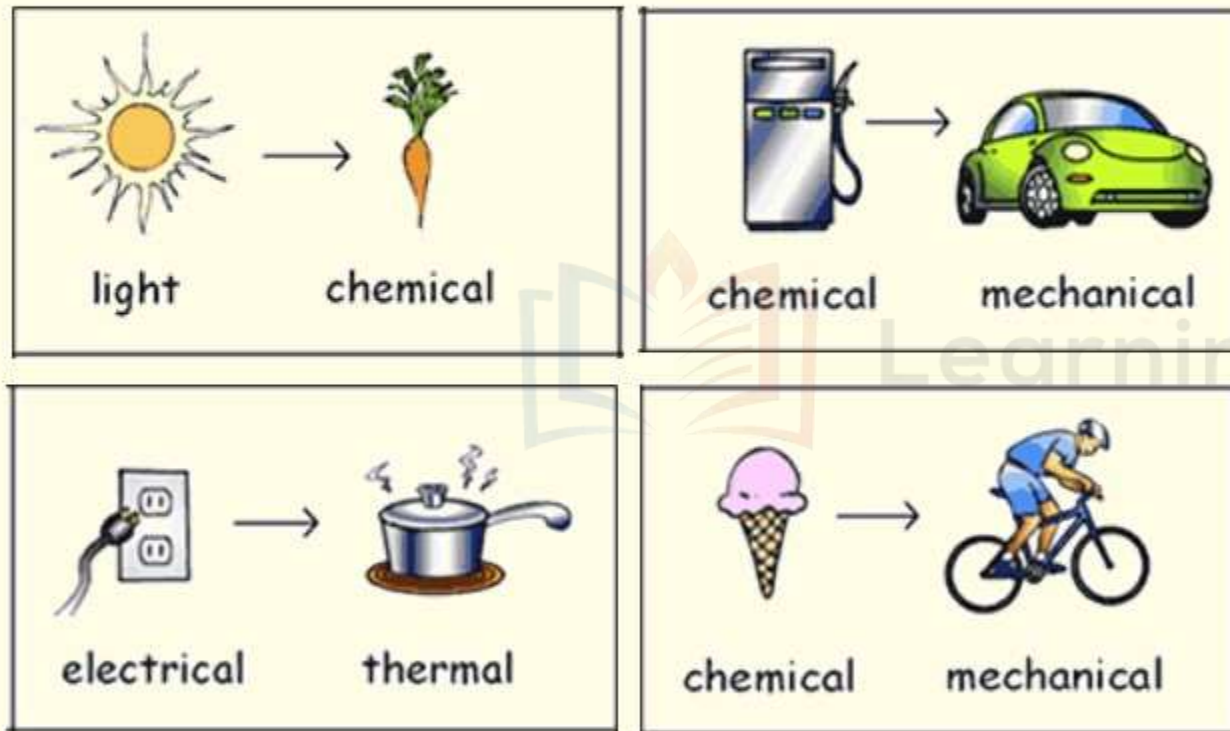
- a) Increases 4 folds
- b) Increases 2 folds
- c) Remains the same
- d) Becomes doubled



HOW THE ENERGY OF THE MOVING PENDULUM CHANGES?



LAW OF CONSERVATION OF ENERGY



- It states that total energy always remains constant in the process.
- It may neither be created nor destroyed.

Energy is interconvertible from one form to another

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

Amrita Naidu

D. Sunil Kumar

L. V. Krishna

D. Chaitanya

K. Srikanth

G. Ravi Babu

D. Joji

K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 -2 4 4 4 4 7 2

0 9 8 4 9 9 4 2 2 9 9